

Referee Report on Manuscript FEJM-D-25-00441

Summary

This paper studies an optimal execution problem in which an agent can trade via three channels: limit orders (LO), market orders (MO), and an internal market maker (MM). A key modeling feature is that trades executed through the internal MM do not generate market impact, in contrast to interbank trades. The problem is formulated as a combined stochastic control and impulse control problem and characterized via a quasi-variational inequality (QVI), and numerical experiments are provided to illustrate the resulting optimal strategies. The topic is relevant and timely. The role of internalization and internal liquidity has become increasingly important in practice, and incorporating these features into optimal execution models is a natural and meaningful extension of the classical LO/MO framework.

Overall Assessment

The paper proposes a practically motivated extension of existing optimal execution models and presents intuitive numerical results. The modeling framework is clear and tractable, and the paper addresses a topic of interest to both academics and practitioners. However, in its current form, the manuscript has several weaknesses in terms of literature positioning, theoretical justification, numerical validation, and presentation. In particular, some claims are stated too strongly relative to the level of justification provided. I therefore recommend the author to prepare a suitable revision answering the following comments:

Major Comments

1. Literature Review

The introduction provides a clear motivation but focuses primarily on classical references (e.g., Almgren–Chriss, Avellaneda–Stoikov, Cartea–Jaimungal). The positioning of the paper relative to more recent work on internalization, internal liquidity, and hybrid execution/market-making models remains limited. The authors should strengthen the literature review and clarify more precisely what is new in their contribution, ideally distinguishing between incremental modeling extensions and genuinely novel aspects.

2. Mathematical Rigor and Model Specification

While the modeling framework is intuitive and practically motivated, several aspects of the mathematical formulation would benefit from clarification and improved internal consistency. The following points aim to indicate concrete directions for revision.

- **General framework (Section 3.1) and consistency with Section 3.2.** Section 3.1 is intended to provide a general combined stochastic control framework (i.e., regular and impulse control), but in its current form it appears to rely on Poisson processes with fixed intensity $\lambda^v \in \mathbb{R}_{>0}^{d_2}$. In contrast, the model in Section 3.2 involves stochastic intensities that depend explicitly

on the control variables. For consistency, Section 3.1 should be reformulated so as to encompass the model of Section 3.2. A natural approach would be to introduce counting processes via their compensators, allowing for control-dependent intensities. For example, one may fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ and explicitly define the counting processes N^k as $\mathbb{Z}_{\geq 0}$ -valued cadlag adapted non-decreasing processes with unit jumps, together with their compensators, i.e., such that

$$N_t^k - \int_0^t \lambda_s^k ds$$

is an $\mathcal{F}_t$ -martingale, where $\lambda_t^k = \lambda^k(u_t)$ with an $\mathcal{F}_t$ -predictable control $(u_t)_{t \geq 0}$. (Or, one can start with Poisson random measures to construct counting processes which are control dependent.) At a minimum, the manuscript should explicitly indicate which existing results are being invoked. In particular, if the authors rely on Seydel, it would be helpful to cite the corresponding results more precisely.

- **Viscosity solution characterization.**

The manuscript states that the value function is the unique viscosity solution of the associated HJBQVI. While such results are standard, they typically rely on structural assumptions (e.g., continuity of coefficients, suitable growth conditions, and a comparison principle). These assumptions are not explicitly verified in the present setting. The authors are encouraged either to state these assumptions explicitly or to moderate the claim accordingly.

- **Verification and admissible controls.**

The link between the HJBQVI and the optimal control strategy is not discussed. It would strengthen the paper to include a brief justification (or appropriate references) explaining how optimal controls are obtained from the solution of the HJBQVI. In this context, the class of admissible controls should also be clarified. In particular, it would be helpful to state that the controls are predictable with respect to a given filtration, and to indicate any implicit constraints (e.g., boundedness or compactness of the control set), which are typically required for dynamic programming and verification arguments. While a full verification theorem may not be necessary, it would strengthen the paper to include a brief discussion or appropriate references (e.g., Fleming and Soner; Øksendal and Sulem) to justify the connection between the solution of the HJBQVI and optimality.

- **Filtration and probabilistic structure.**

The underlying filtration is not explicitly specified. Since the model involves control-dependent arrival intensities, it is important to fix the filtration independently of the control and to define the counting processes via their compensators with respect to this filtration. Clarifying this point would remove potential ambiguity and ensure a mathematically consistent formulation. (See the above general framework and consistency part.)

- **Specification of control ranges.**

The admissible ranges of the control variables (in particular the spread con-

trols δ^L and δ^I) are not explicitly specified. Are these controls constrained to a bounded or compact set, which is often assumed in the theoretical analysis? An appropriate choice of admissible control variables is sometimes necessary to show the verification theorem stated above.

- **Inventory constraints and boundary behavior.**

The model appears to assume that the inventory remains nonnegative. This is explicit for market orders (through $\zeta \in \{0, 1, \dots, q\}$), but it would be helpful to clarify how the boundary case $q = 0$ is handled for limit-order and internal-MM executions (e.g., by setting the corresponding intensities to zero or by restricting admissible controls accordingly).

Addressing these points would significantly improve the mathematical rigor, internal consistency, and clarity of the manuscript, without altering its main contributions.

3. Numerical results: lack of quantitative comparison

The numerical section provides useful qualitative insights into the structure of the optimal strategies. However, it does not provide a clear quantitative assessment of the benefits associated with the introduction of the internal MM channel. Further numerical results are helpful for readers to directly quantify the incremental value of internal liquidity. More specifically, the authors are encouraged to include:

- a comparison of the value functions, $H(0, x, s, q)(= x + sq + h(0, q))$ or $h(0, q)$, for the two strategies, that is, LO/MO and LO/MO/MM;
- a sensitivity analysis with respect to key parameters, such as the internal liquidity intensity λ_I, κ_I impact coefficients α_M, α , and risk aversion ϕ , for example.

These additions would substantially strengthen the empirical relevance of the numerical results and make the contribution of the internal MM channel more transparent.

Minor Comments

1. p.2, l.39: Use “This section” instead of “This chapter”.
2. In Section 3.4, when we treat HJB equation, it is better to use $(\delta_t^{I*}(q), \delta_t^{L*}(q))$ for the maximizers instead of $\delta_t^{I*}, \delta_t^{L*}$.